

Latency Validation Method Review

Context-Aware Digital Twin for Polymer-Joint Cobots

DECISION	Yes - the project can move on. The current latency methodology is technically sound for the present research stage because it separates communication delay from robot motion-response delay and uses cautious terminology. It is not yet a claim of final validated latency performance; the final experiment still needs a few protocol details frozen.
----------	--

Why this is enough to move forward

The saved roadmap already treats the latency method as established infrastructure: CRI ACK round-trip time (RTT) is one dataset, while telemetry-observed motion response is a second dataset. The roadmap also explicitly says not to spend more effort optimizing the latency pipeline unless a concrete experimental problem appears. Its immediate research priority is model identification and the later adaptive/context-aware logic.

Project rule	Meaning
Keep network and motion-response measurements separate	Prevents a network RTT number from being misinterpreted as robot/actuator dynamics.
RTT/2 is only an estimated one-way delay	Forward and reverse paths may not be identical; direct one-way timing needs synchronized clocks.
TX-to-Motion is not pure actuator latency	It contains robot processing, control-cycle timing, physical response, telemetry return/update effects, and possibly other delays.
Use actual timestamps and measured dt	The checked telemetry log is non-uniformly sampled; nominal sampling rate must not replace measured time spacing.

NEXT STEP	Proceed to the identification pipeline: confirm the experiment logger/dataset, collect clean no-payload/no-contact trajectories, preprocess with measured dt, fit Model A, compare Model B with acceleration, and freeze the final regressor before implementing VFF-RLS and the context state machine.
-----------	---

1. Current latency method - simple explanation

Think of the project as measuring two different questions. Method 1 asks how long communication takes. Method 2 asks how long it takes before the commanded physical motion is visible in returned robot telemetry. Keeping these questions separate is the main strength of the current method.

Method 1 - CRI application-layer ACK round-trip time



$$\text{RTT} = (t_{\text{ack_rx}} - t_{\text{tx}}) \times 1000 \text{ ms}$$

When the PC sends a CRI command, record the transmit timestamp t_{tx} . When the related acknowledgement returns, record $t_{\text{ack_rx}}$. The same PC clock measures both ends of the interval, which makes RTT practical and avoids the cross-device clock-synchronization problem of direct one-way timing.

$$\text{RTT_ms} = (t_{\text{ack_rx}} - t_{\text{tx}}) \times 1000$$

If a one-way number is useful for discussion, the project may calculate:

$$\text{Estimated_one_way_ms} = \text{RTT_ms} / 2$$

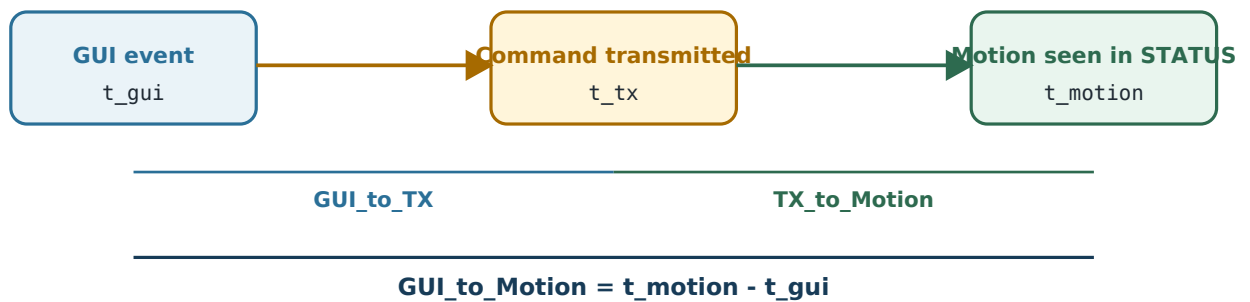
Wording rule

Call $\text{RTT}/2$ an estimated one-way communication delay under an approximate symmetry assumption. Do not call it a measured one-way latency.

Example

If $t_{\text{tx}} = 12.500 \text{ s}$ and $t_{\text{ack_rx}} = 12.518 \text{ s}$, then $\text{RTT} = 18 \text{ ms}$. $\text{RTT}/2 = 9 \text{ ms}$ may be reported only as an estimated one-way communication delay.

2. Telemetry-observed motion response



This experiment follows the command from the GUI to transmission and then waits until motion is first detectable in incoming CRI STATUS telemetry. It therefore measures an application/robot response pathway, not just packet transport.

```

GUI_to_TX_ms = (t_tx - t_gui) x 1000
TX_to_Motion_ms = (t_motion - t_tx) x 1000
GUI_to_Motion_ms = (t_motion - t_gui) x 1000
  
```

For a matched event, GUI_to_Motion is the sum of GUI_to_TX and TX_to_Motion. The project should still avoid calling TX_to_Motion "actuator latency" because the measured interval contains more than the actuator itself.

Safe statement	Statement to avoid
"CRI ACK RTT = 18 ms"	"Network one-way latency = 9 ms" without qualification
"Estimated one-way delay = 9 ms, assuming approximate symmetry"	"RTT/2 is a measured one-way delay"
"TX-to-Motion = 35 ms, with motion observed in STATUS telemetry"	"Actuator latency = 35 ms"
"RTT and motion response are independent datasets unless explicitly paired"	Subtracting average RTT/2 from average TX-to-Motion to invent an actuator-only delay

3. Is the method scientifically defensible?

Yes, for the current project stage. The method follows common timing-measurement principles: measure intervals using explicit timestamps, use RTT when one clock can observe both ends, distinguish one-way delay from RTT, and distinguish communication timing from robot response timing. Comparable robotics studies also separate communication delay from command/response or tracking delay.

Aspect	Status	Reason
RTT definition	Strong	Standard round-trip timing concept; same source clock observes send and return.
RTT/2 interpretation	Strong	Project correctly labels it an estimate, not direct one-way measurement.
Motion-response timing	Strong concept	Command timestamp is related to physical response visible in robot telemetry.
Motion-onset criterion	Still to freeze	The saved project does not yet define the exact $q / \dot{q}$ threshold or persistence rule used to declare t_{motion} .
Final statistics	Still to freeze	The saved method defines per-sample equations but not a mandatory final summary statistic set.

Small items to freeze before the final paper experiment

- Clock: use a monotonic/high-resolution clock for intervals; document which process/machine timestamps each event.
- Event matching: attach a command/sequence ID so an ACK and a motion-response sample can be associated with the correct command when correlation is intended.
- Motion onset: define one deterministic rule for t_{motion} (for example a joint-velocity or position-change threshold plus persistence) and freeze it before final tests.
- Replicates: collect enough repeated trials under fixed conditions; preserve raw per-trial values rather than only an average.
- Statistics: as a recommended extension, report mean, median, standard deviation, P95, P99 and max for each latency dataset. P95 is common in ROS latency work, but this exact list is a recommendation rather than a currently frozen project requirement.
- Raw telemetry: keep original timestamps and signals immutable; compute derived latency fields in additional columns/files.

Bottom line	These refinements are protocol-completion items, not blockers. They can be finalized when preparing the final latency experiment. The research effort should now return to the model-identification milestone.
--------------------	--

4. References equivalent or closely analogous to the method

The references below are not identical to the igus ReBeL/CRI setup. They are methodologically equivalent or closely analogous: they support RTT measurement, the caution required for one-way delay, timestamp-based ROS/robot latency measurement, and separation of communication timing from physical command-response timing.

Reference	Closest project element	Why it is useful
RFC 2681	CRI ACK RTT	Formal round-trip delay metric using send and return observations; explains why RTT is convenient without source/destination clock synchronization.
RFC 7679	RTT/2 caution	Direct one-way delay requires closely synchronized source/destination clocks and can differ by direction because paths may be asymmetric.
Lim et al., 2026	Method 1 - timestamped ACK return	ROS 2 experiment uses <code>t_send</code> , <code>t_rcv</code> and <code>t_return</code> ; computes one-way and RTT, and treats RTT as the primary cross-host reference because send/return are on one clock.
Kronauer et al., 2021	ROS end-to-end latency / statistics	Defines latency from publish to subscriber callback and uses the 95th percentile as a practical timing statistic.
Hummel et al., 2019	Separate network vs command-response timing	Distributed drone study evaluates network RTT separately from command response time, closely matching the project philosophy.
Baklouti et al., 2021	Robot motion response / tracking delay	ROS-controlled industrial robot study evaluates response time and tracking delay as robot/control performance rather than calling it pure network delay.

Reference list

- [R1] Almes, G., Kalidindi, S., & Zekauskas, M. (1999). A Round-trip Delay Metric for IPPM. RFC 2681. RFC Editor. <https://www.rfc-editor.org/rfc/rfc2681.html>
- [R2] Almes, G., Kalidindi, S., Zekauskas, M., & Morton, A. (Ed.). (2016). A One-Way Delay Metric for IP Performance Metrics (IPPM). RFC 7679 / STD 81. RFC Editor. <https://www.rfc-editor.org/rfc/rfc7679.html>
- [R3] Lim, J. M., Kim, K. W., Choi, B. W., & Delgado, R. (2026). A Docker-Enabled Real-Time Framework for Robotic Applications in Heterogeneous ROS 2 Environments. *Processes*, 14(5), 804. <https://doi.org/10.3390/pr14050804>
- [R4] Kronauer, T., Pohlmann, J., Matthe, M., Smejkal, T., & Fettweis, G. P. (2021). Latency Analysis of ROS2 Multi-Node Systems. 2021 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI), 1-7. <https://doi.org/10.1109/MFI52462.2021.9591166>
- [R5] Hummel, K. A., Pollak, M., & Krahofer, J. (2019). A Distributed Architecture for Human-Drone Teaming: Timing Challenges and Interaction Opportunities. *Sensors*, 19(6), 1379. <https://doi.org/10.3390/s19061379>
- [R6] Baklouti, S., Gallot, G., Viaud, J., & Subrin, K. (2021). On the Improvement of ROS-Based Control for Teleoperated Yaskawa Robots. *Applied Sciences*, 11(16), 7190. <https://doi.org/10.3390/app11167190>

5. Recommended project continuation

Because the latency methodology is sufficiently mature for now, the next work should follow the saved roadmap rather than redesigning the latency pipeline.

Order	Action	Purpose
1	Confirm dedicated identification logger / CSV	Raw time, q, qdot, current, run metadata; verify missing values and timestamp spacing.
2	Collect normal-operation data	No payload, no intentional contact/collision; diverse single- and multi-joint trajectories and speeds.
3	Offline preprocessing	Use measured dt; filter qdot; construct candidate qddot; inspect data quality.
4	Fit Model A	Baseline regressor without acceleration; train/held-out validation.
5	Fit Model B	Add acceleration only if it improves held-out prediction without unstable/noisy parameters.
6	Freeze final regressor	Only after this move to context-gated VFF-RLS, state-machine logic, calibration, and final validation.

Research checkpoint	Do not interpret "move on" as "latency is finished forever." It means the method is good enough to stop consuming research time now. Before final manuscript results, freeze the motion-onset rule, run the final repeated latency experiments, and report their distributions with the method unchanged.
----------------------------	---

Project-source basis

This review is grounded in the saved project files dated 25 Sep 2026: Context-Aware Digital Twin Roadmap and PROJECT: Context-Aware Digital Twin for Polymer-Joint Cobots. Those files define the two-method latency approach, require separate RTT and motion-response datasets unless explicitly correlated, caution against calling RTT/2 a measured one-way delay or TX-to-Motion pure actuator latency, and direct the project to move next into model identification rather than further GUI/latency optimization.

Prepared for project continuation - 28 Sep 2026.